

Mini Project

CSE405 (Computer Networks)

Title: Design a full-fledged network for the University of Scholars with multiple subnets.

Dynamics

The project must be done individually. A technical report consisting of the design criteria and explanation must be submitted upon the completion of your work. Please submit both packet tracer file (.pkt) and the report (.pdf) using Google classroom.

Background

University of Scholars, is an enterprise like East West University, owns many computers, with a complex network infrastructure. Apart from wired internet access to all the classrooms, labs, employee PCs, library and other administrative and academic wings, the university also provides wireless internet access for every campus. On top of that, the university runs complex networked systems to support several of its business processes like admissions, advising, results, eTender, library management, accounts and so on.

This complex network infrastructure is subnetted and switching/routing mechanisms are in practice.

Tasks

Your task is to create a complete model of a complex network by discovering the interconnectivity of the systems and subnetworks, which will reflect the University's structure and facilities. Features within the network will include the following:

- The web page of the university will reflect its own information.
- A **single DNS** server needs to be installed to locate webserver - meaning people will browse University's web site from any campus with the following address: <http://www.scholars.edu.bd>
- Configure the whole network in such a way that IP for the hosts of different campuses will be automatically assigned by a **single DHCP server**. If a single DHCP is not doable by you, then use multiple DHCP servers; however, that will be discredited. IP pool for the DHCP server will be '**A**' class **ONLY**.

- Create five LANs for campus 1, campus 2, campus 3, campus 4 and campus 5; among the hosts in a network make sure some wireless hosts are added in addition to wired hosts.
- University's full network has covered its five campuses with five routers; connections between the campus routers i.e. the topology of the network is given at the end of the handout; you must (strictly) follow the given topology.
- Connectivity between all the hosts needs to be established.
- For routing protocol please use dynamic algorithm only i.e., OSPF.
- Connection between all the routers will be serial connections; if serial connections are not doable, use straight through or cross-over connections; however, that will be discredited.

While designing, keep the issue of future expansion/growth in mind for each of the subnets (if required) and preserve spaces. In physical design, it is a good practice to have a server room where all the servers are in one LAN segment.

Tools

You are free to use any tools necessary. For the design of network model and for implementation of servers 'Packet tracer' is recommended.

Report

Develop documentation professionally where requirements and design specifications will be included.

- Include the introduction, purpose of the network and design issues (if any)
- Limitations and lines of code used to configure the network
- Routing tables of all the routers
- Please include the complete diagram of the network including hosts and servers
- Description on the present network and how the campus can be scaled up in the future
- Conclusion

Assessment methods:

You will be assessed depending on functionality of the network meeting all the requirements. In addition, your report should reflect professionalism and network design should be concise and well organized.

Grading

The total marks for the project is 10. Assessment would be on complex professional design and arrangement of the network that meet all requirements with functionality and on the report.

Submission Process:

You should submit your packet tracer file (.pkt) and the report (.pdf) using Google classroom within the deadline, mailing is not an option. Not meeting the deadline will be discredited; however, submission after the assessment is over has no value.

Special note: Keep your own design with yourself; do NOT distribute your design. Resemblance will be penalized.

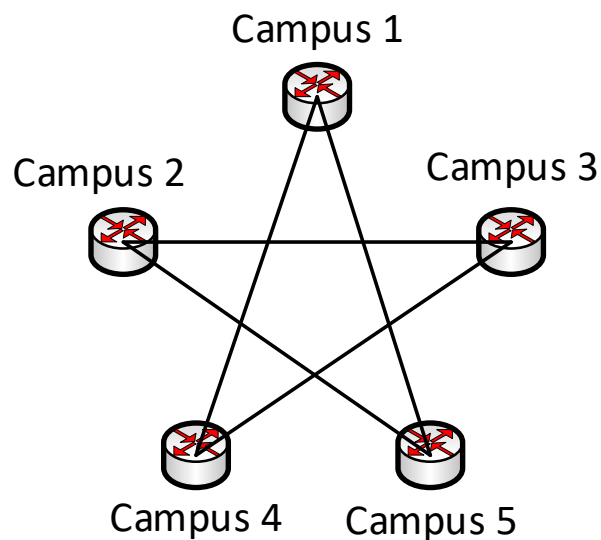


Figure 1: Physical connectivity of the Campus Network